

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 13

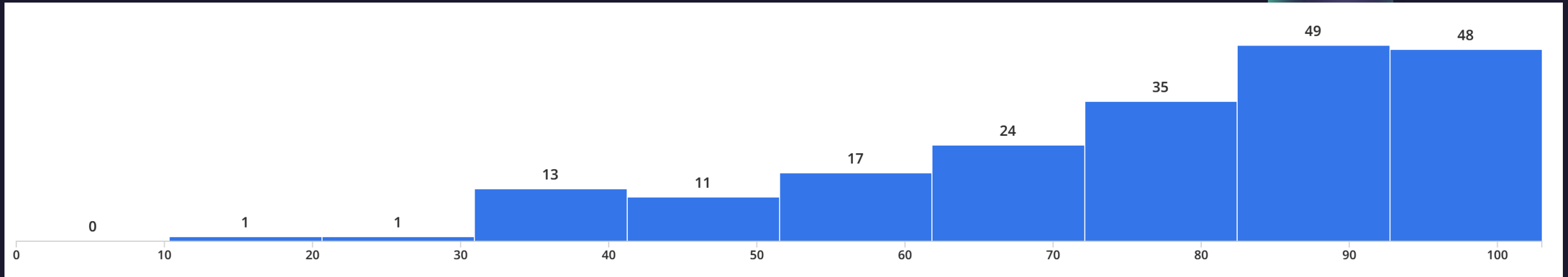


Any Reminders? Any Questions?

- I have regular office hours Mondays via Webex 1-2pm
- I have regular office hours Thursdays in-person 3:30-4:30pm
- Calc 3 Calc Night: MONT 104 at 6:30-8:30pm on Thursdays!
- Quiz on Friday on polar coordinates and beginning of triple integrals

EXAM 1

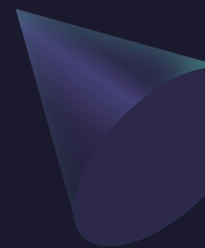
- Average = 77, Median = 81



- Exam I – One-Page Correction (+5)



ALVARO: Start the recording!



“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

More on Triple Integrals

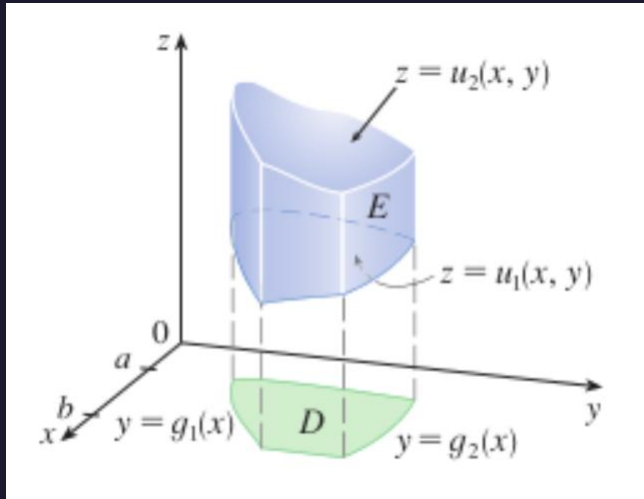




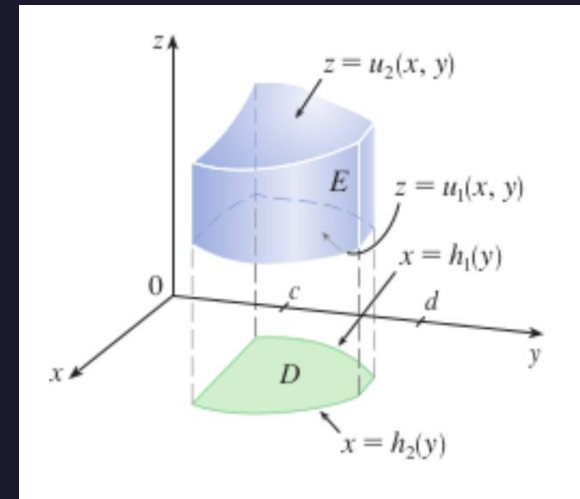
Today – Triple Integrals

- Triple Integrals in Boxes
- Triple Integrals in Regions
- Changing the Order of Integration

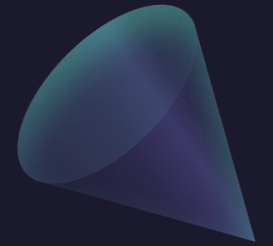
Triple Integrals in General Regions: Type I, ...



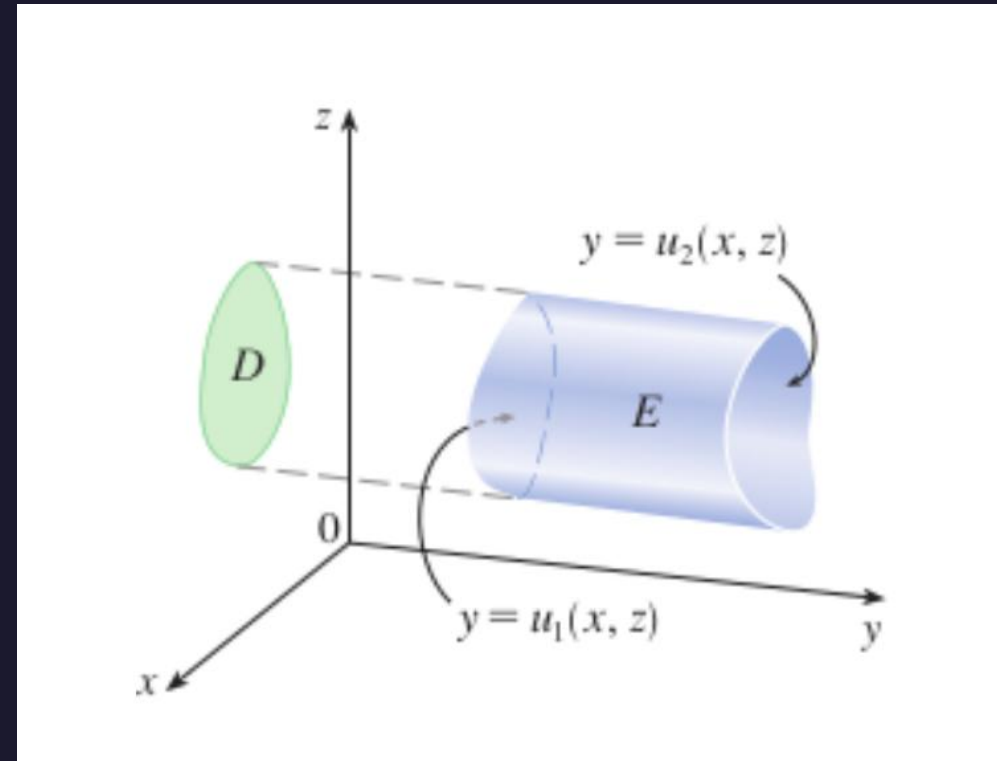
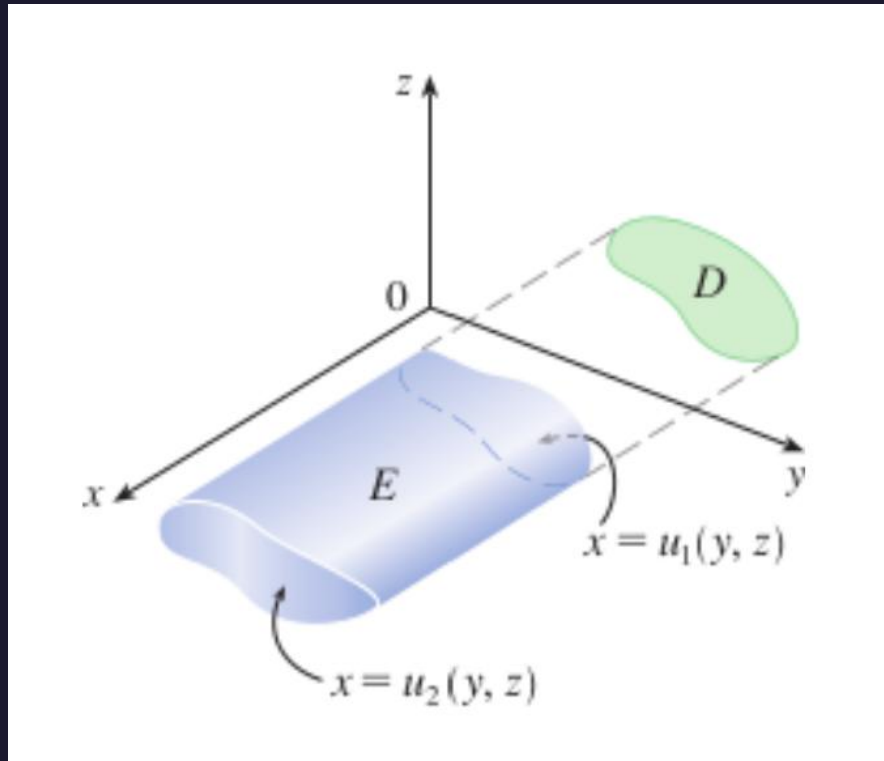
$$\iiint_E f(x, y, z) dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz dx dy$$



$$\iiint_E f(x, y, z) dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz dx dy$$



Triple Integrals in General Regions: Type II and Type III



Example: Evaluate the following triple integral

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) dz dy dx$$

$$= \int_0^1 \int_0^{x^2} \left. xz + yz + \frac{z^2}{2} \right|_0^y dy dx$$

$$= \int_0^1 \int_0^{x^2} \left(xy + y^2 + \frac{y^2}{2} \right) dy dx = \int_0^1 \int_0^{x^2} \left(xy + \frac{3}{2} y^2 \right) dy dx$$

$$= \int_0^1 \left. \frac{xy^2}{2} + \frac{y^3}{2} \right|_0^{x^2} dx = \int_0^1 \left(\frac{x^5}{2} + \frac{x^6}{2} \right) dx$$

$$= \left. \frac{x^6}{12} + \frac{x^7}{14} \right|_0^1 = \frac{1}{12} + \frac{1}{14} = \boxed{\frac{13}{84}}$$

Example: Evaluate the following triple integral

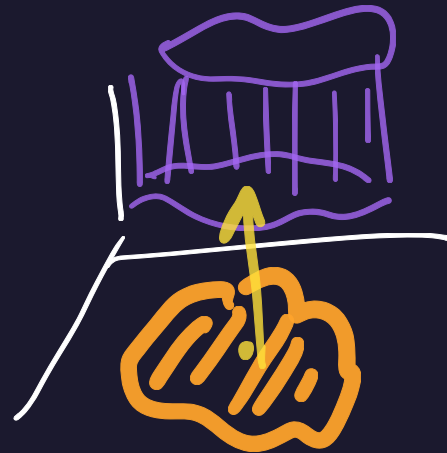
$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) dz dy dx$$

$$E = \left\{ \begin{array}{l} 0 \leq x \leq 1 \\ 0 \leq y \leq x^2 \\ 0 \leq z \leq y \end{array} \right\} \quad \begin{array}{l} \text{TYPE I} \\ xy\text{-plane} \end{array}$$



TYPE II
yz plane



Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) dz dy dx$$

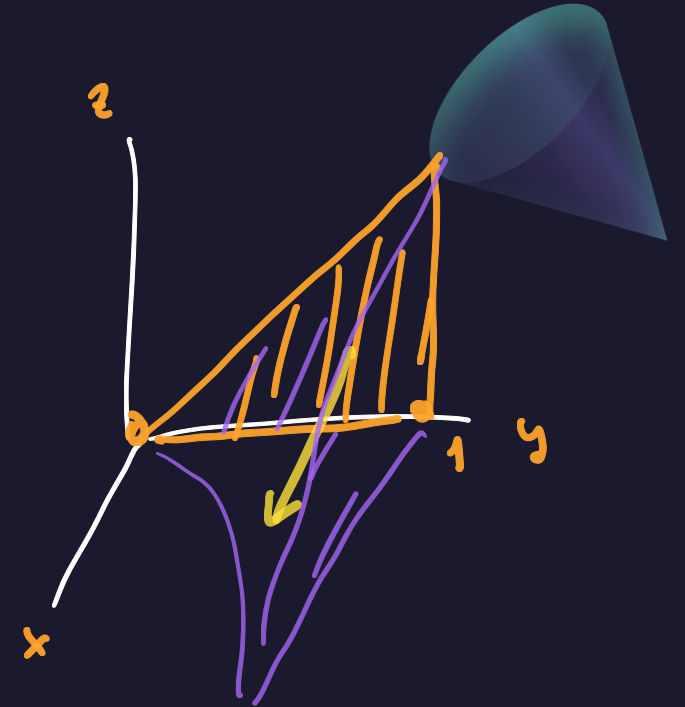
TYPE I

$$\begin{cases} 0 \leq x \leq 1 \\ 0 \leq y \leq x^2 \leq 1 \\ 0 \leq z \leq y \leq 1 \end{cases}$$

TYPE II

$$\begin{cases} 0 \leq y \leq 1 \\ 0 \leq z \leq y \\ \sqrt{y} \leq x \leq 1 \end{cases}$$

$$\int_0^1 \int_0^y \int_{\sqrt{y}}^1 (x + y + z) dx dz dy$$



Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) dz dy dx$$

TYPE I

$$\begin{cases} 0 \leq x \leq 1 \\ 0 \leq y \leq x^2 \leq 1 \\ 0 \leq z \leq y \end{cases}$$

$\rightarrow 0 \leq z \leq y \leq x^2$
 $\rightarrow z \leq y \leq x^2$

TYPE III

(xz plane)

$$\begin{cases} 0 \leq x \leq 1 \\ 0 \leq z \leq x^2 \end{cases} \quad \text{ } x^2$$

$z \leq y \leq x^2$

$$\int_0^1 \int_0^{x^2} \int_z^{x^2} (x + y + z) dy dz dx$$

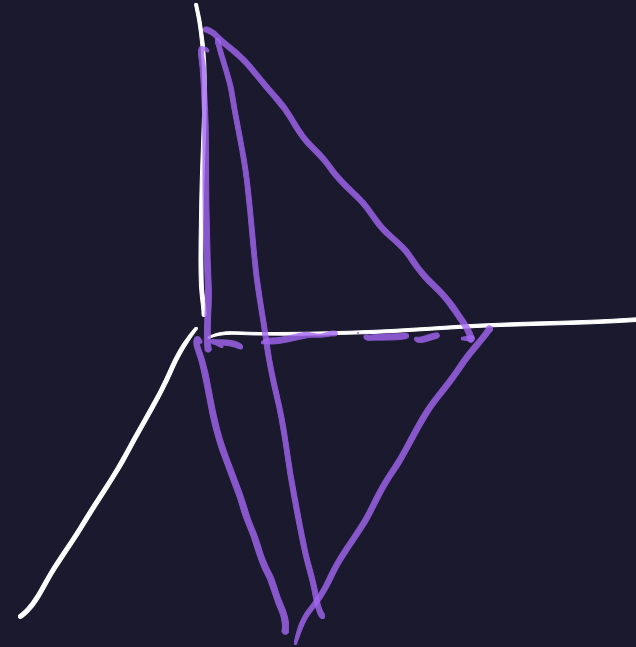
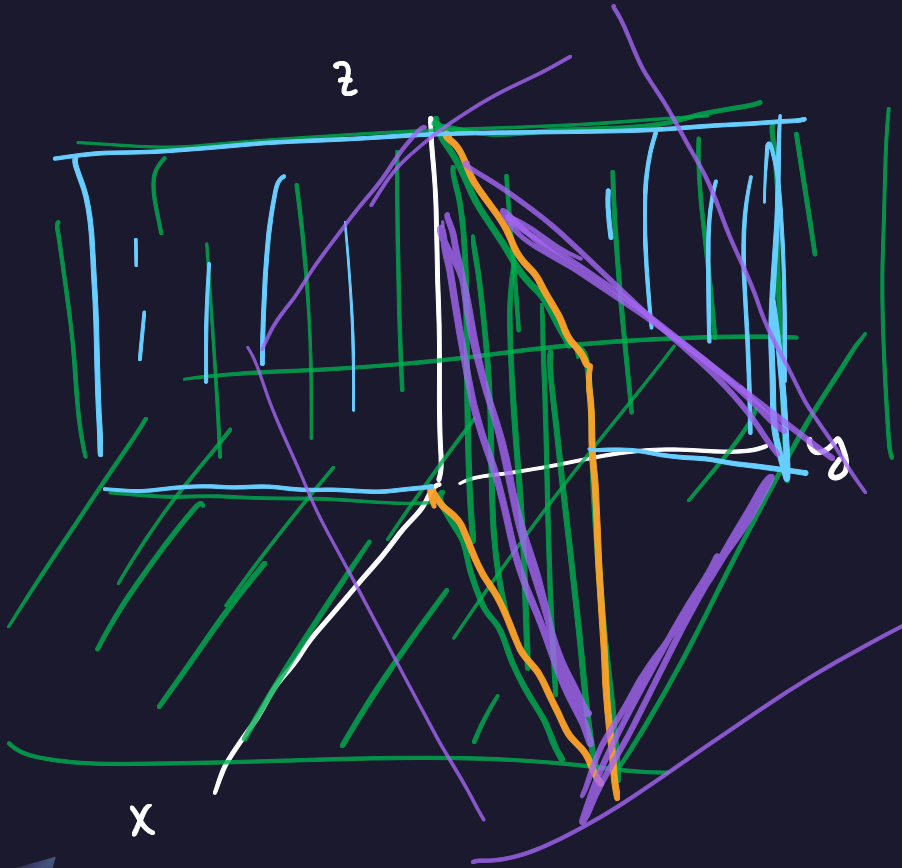
Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

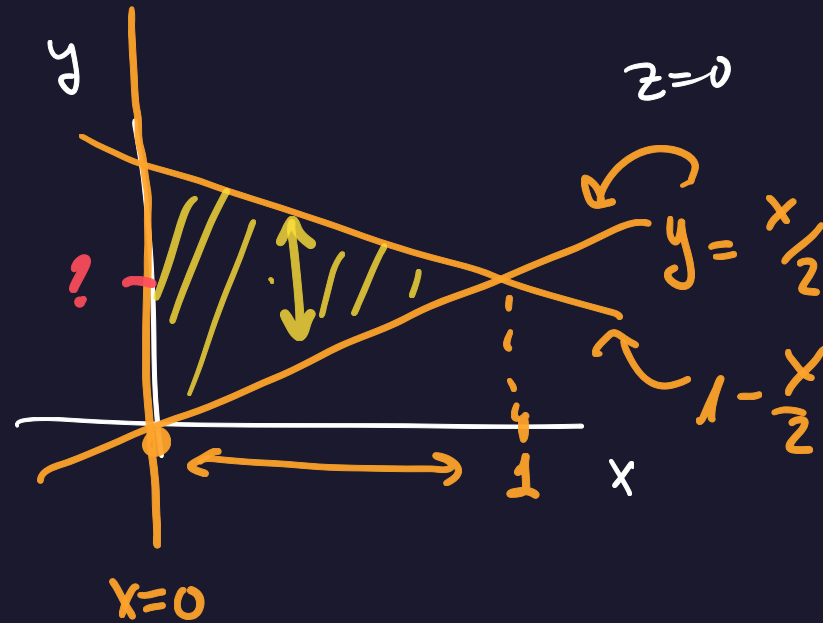
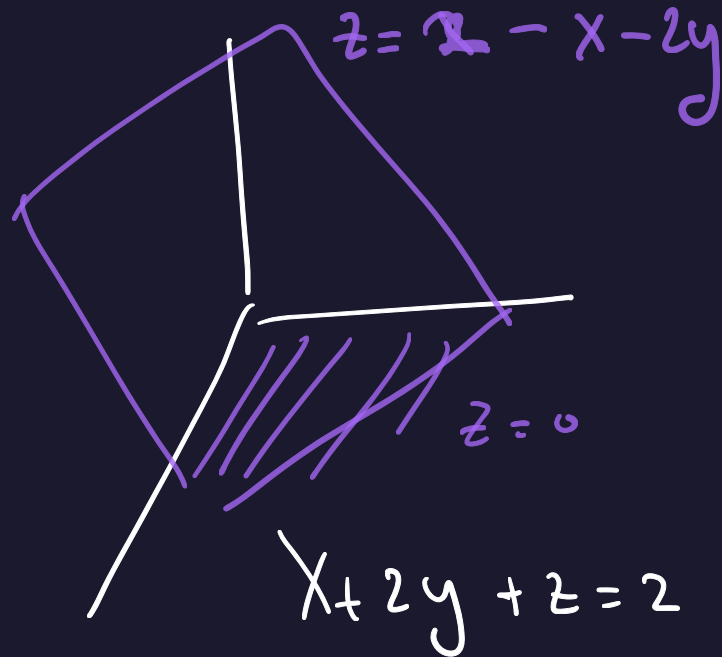
Example: Find the volume of the solid bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$, and $z = 0$.

$$x = 2y \quad y = \frac{x}{2}$$



TYPE 1

Example: Find the volume of the solid bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$, and $z = 0$.



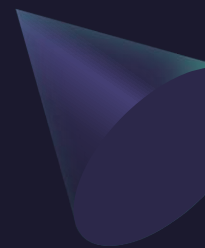
$$\begin{aligned}
 x + 2y + z &= 2 \\
 z &= 0 \\
 x + 2y &= 2 \quad \leadsto \quad y = \frac{2-x}{2} = 1 - \frac{x}{2}
 \end{aligned}
 \quad \left. \begin{aligned} y &= \frac{x}{2} \\ y &= 1 - \frac{x}{2} \end{aligned} \right\} \Rightarrow 1 - \frac{x}{2} = \frac{x}{2} \Rightarrow 1 = x$$

Questions?

$$\int_0^1 \int_{x/2}^{1-x/2} \int_0^{2-x-2y} 1 \, dz \, dy \, dx = V(E)$$




ALVARO: Start the recording!

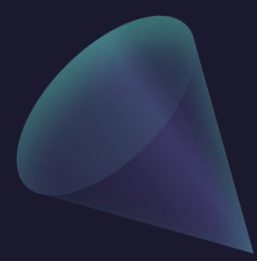


“Calculus 3”

Multi-Variable Calculus

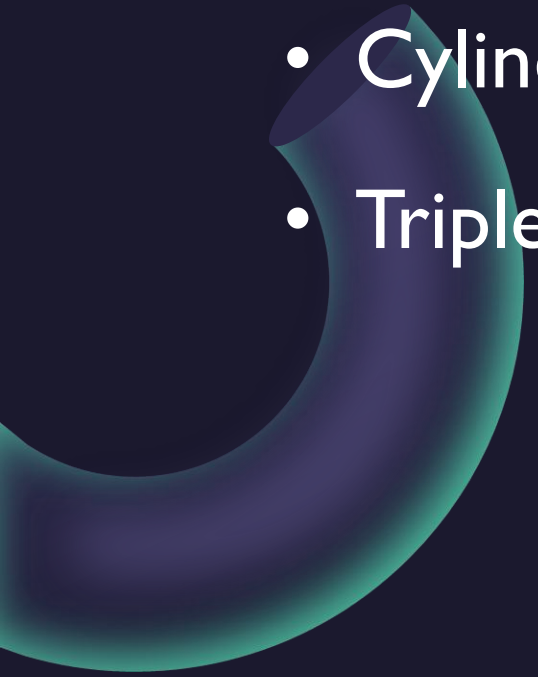
Instructor: Álvaro Lozano-Robledo

Triple Integrals in Cylindrical





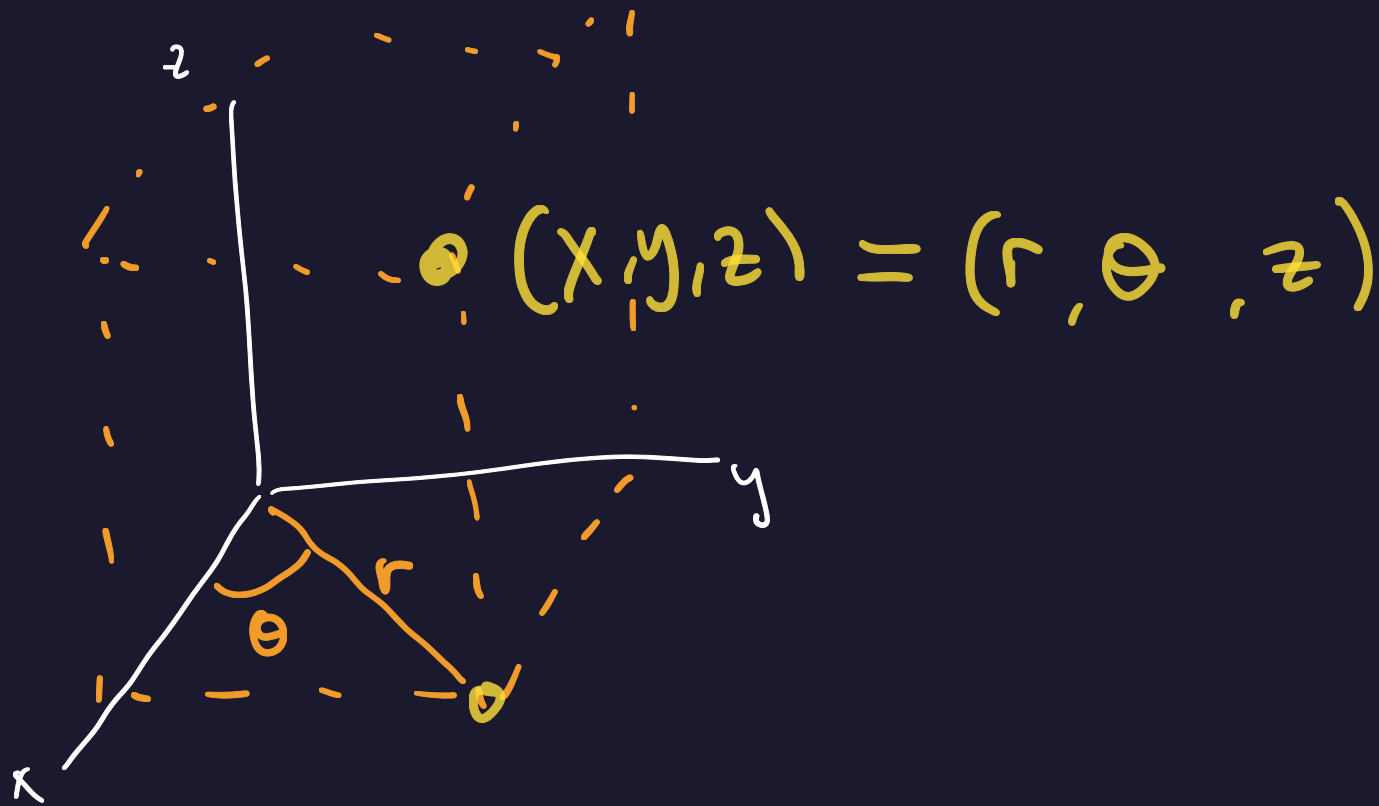
Today – Triple Integrals in Cylindrical Coordinates

- 
- Cylindrical Coordinates
 - Triple Integrals in Cylindrical Coordinates

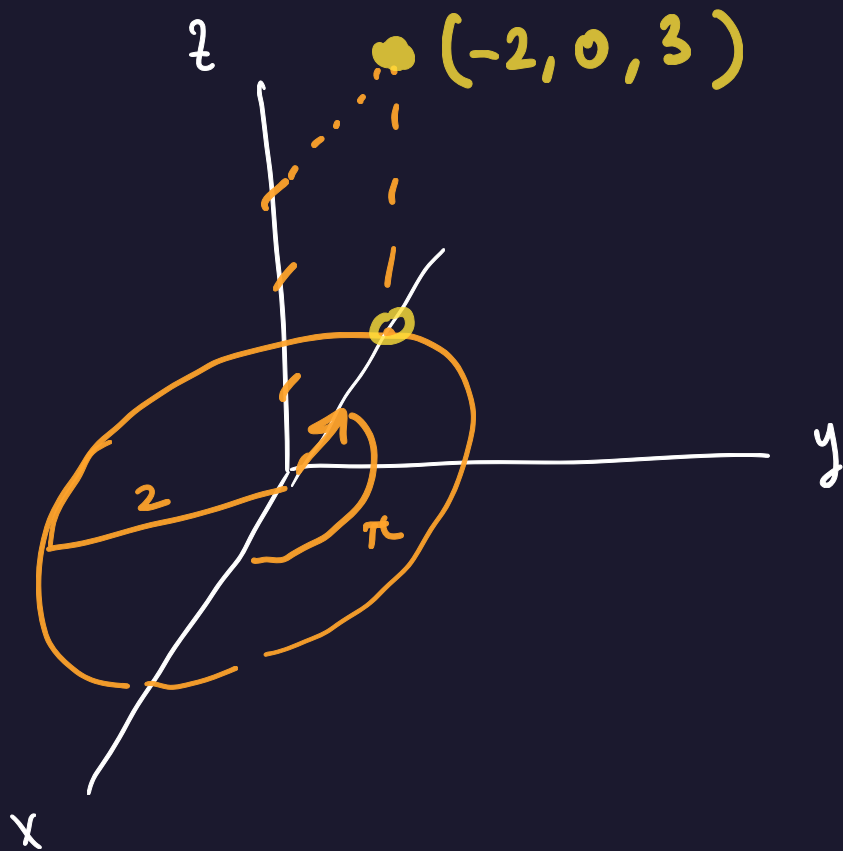
Cylindrical Coordinates

$$x = r \cos \theta \quad y = r \sin \theta \quad z = z$$

$$r^2 = x^2 + y^2 \quad \tan \theta = \frac{y}{x} \quad z = z$$



Example: Plot the point with cylindrical coordinates $(r, \theta, z) = (2, \pi, 3)$

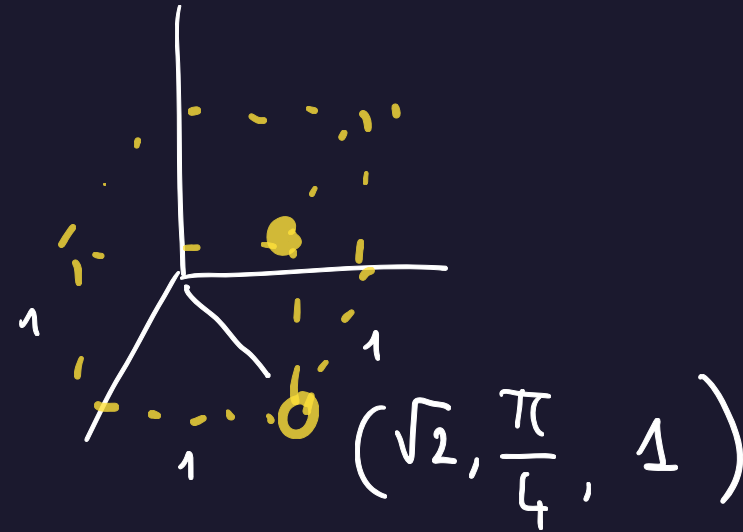
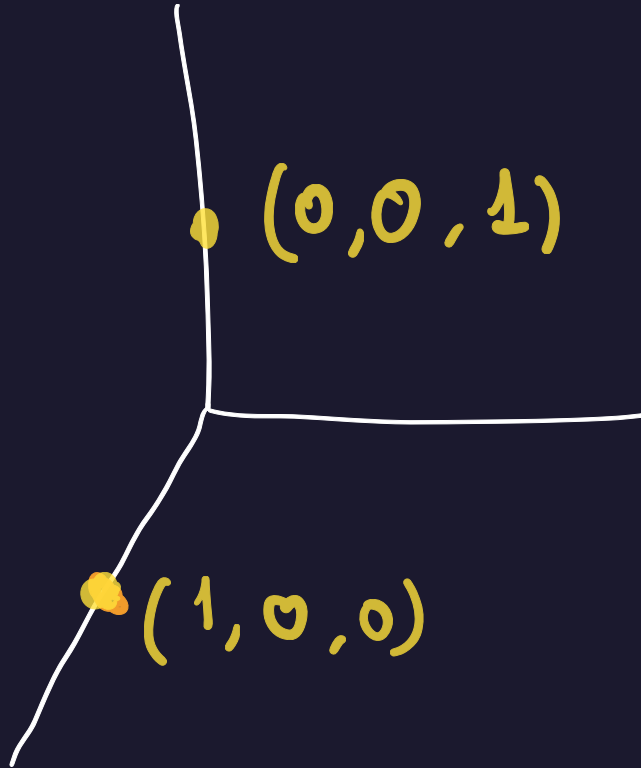


$$(x, y, z) = ?$$

$$\begin{cases} x = r \cos \theta = 2 \cdot (-1) = -2 \\ y = r \sin \theta = 2 \cdot 0 = 0 \\ z = z = 3 \end{cases}$$

$$\underbrace{\hspace{10em}}_{(-2, 0, 3)}$$

Example: Find the cylindrical coordinates of the points $(1,0,0)$, $(0,0,1)$, and $(1,1,1)$.

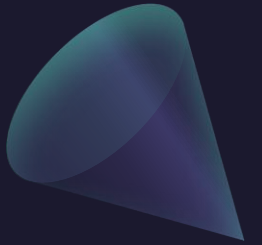


Example: Describe the surface given in cylindrical coordinates by $z = r$.



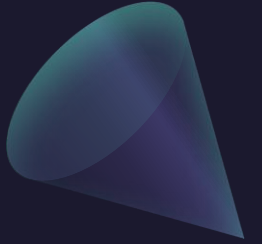
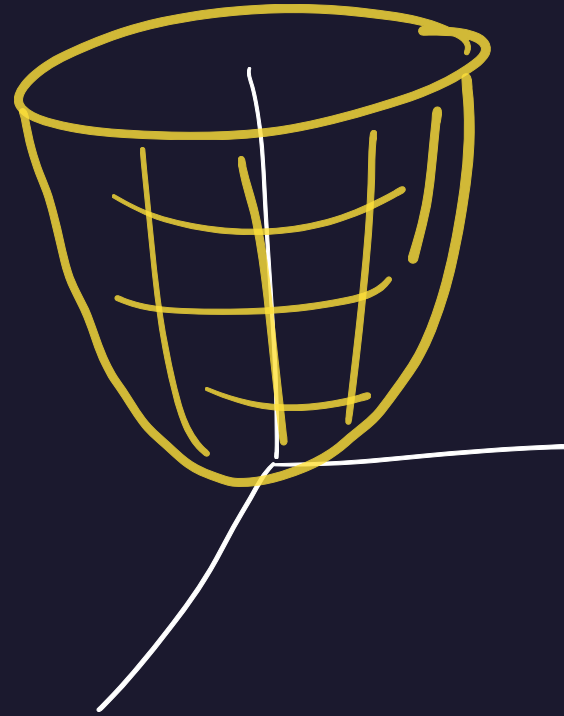
$$z = \sqrt{x^2 + y^2} \quad \leftarrow \text{cone}$$

$$z^2 = x^2 + y^2 \quad \leftarrow \text{double cone}$$

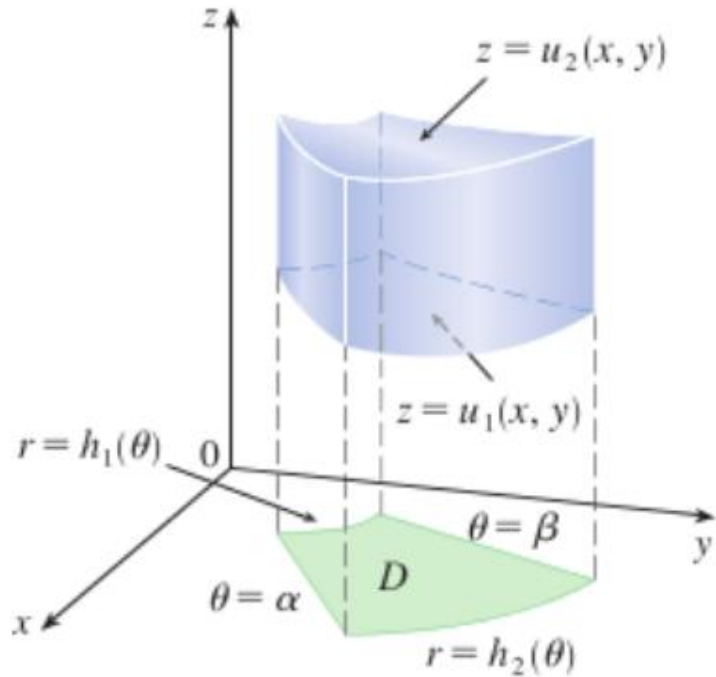


Example: Describe the surface given in cylindrical coordinates by $z = r^2$.

$$z = r^2 = x^2 + y^2$$



Triple Integrals in Cylindrical Coordinates (of Type I)



$$\iiint_E f(x, y, z) \, dV = \iint_D \left[\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) \, dz \right] dA$$

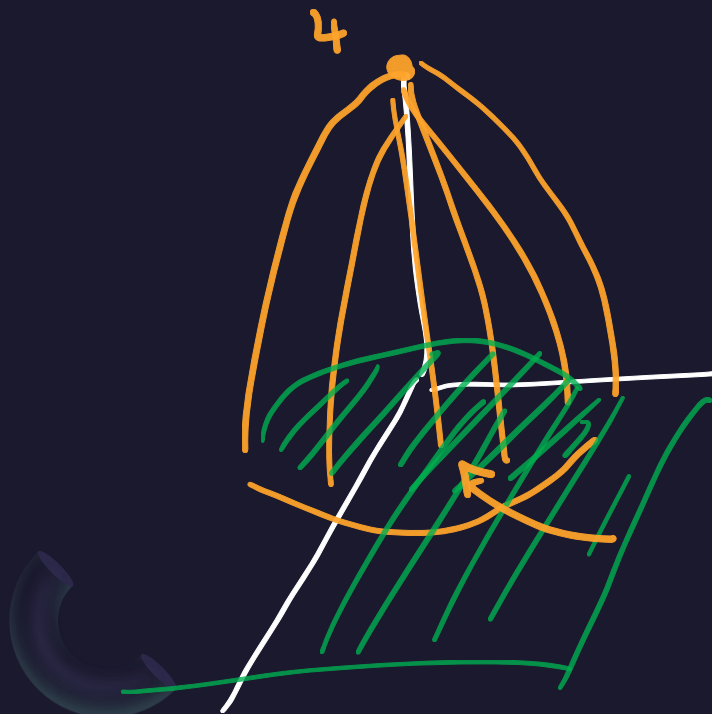
conv. factor !!!

$$\iiint_E f(x, y, z) \, dV = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r \cos \theta, r \sin \theta)}^{u_2(r \cos \theta, r \sin \theta)} f(r \cos \theta, r \sin \theta, z) \, r \, dz \, dr \, d\theta$$

Example: Evaluate the value of the triple integral

$$\iiint_E x^2 dV$$

where E is the solid bounded by the surface $z = 4 - x^2 - y^2$ and above the xy -plane.



$$\left. \begin{array}{l} z = 4 - x^2 - y^2 \\ z = 0 \end{array} \right\}$$

$$x^2 + y^2 = 4$$

$$(r, \theta)$$

$$0 \leq r \leq 2$$

$$0 \leq \theta \leq 2\pi$$

$$\int_0^{2\pi} \int_0^2 \int_0^{4-r^2}$$

$$(r \cos \theta)^2 \cdot r \cdot dz dr d\theta$$

$$4 - x^2 - y^2 = 4 - r^2 \quad r^3 \cos^2 \theta$$



Example: Evaluate the value of the triple integral

$$\iiint_E x^2 dV$$

$$\int_0^{2\pi} \int_0^2 \int_0^{4-r^2} r^3 \cos^2 \theta \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^2 r^3 \cos^2 \theta \, z \Big|_0^{4-r^2} dr \, d\theta$$

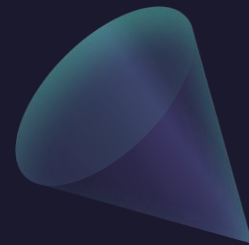
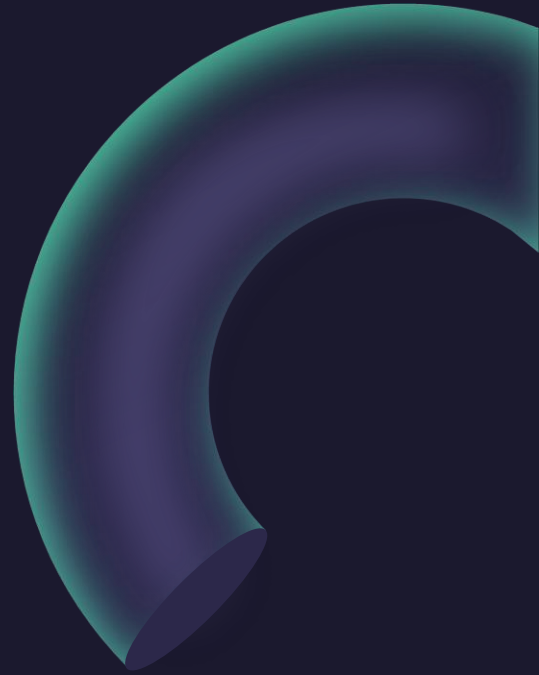
$$= \int_0^{2\pi} \int_0^2 (4-r^2) r^3 \cos^2 \theta \, dr \, d\theta = \int_0^{2\pi} \underbrace{\cos^2 \theta \, d\theta}_{\pi} \cdot \int_0^2 (4r^3 - r^5) \, dr$$

$$= \pi \cdot \left(r^4 - \frac{r^6}{6} \right) \Big|_0^2 = \pi \cdot \left(16 - \frac{32}{3} \right) = \boxed{\frac{16}{3} \pi}$$

$$\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2}$$

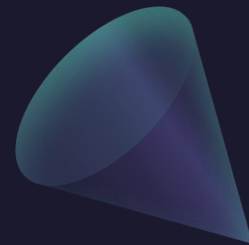
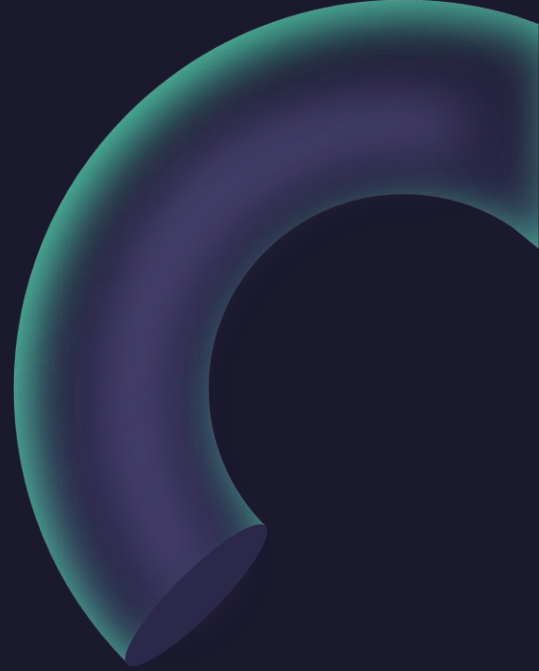
Example: Evaluate the value of the triple integral

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2+y^2) \, dz \, dy \, dx$$



Example: Evaluate the value of the triple integral

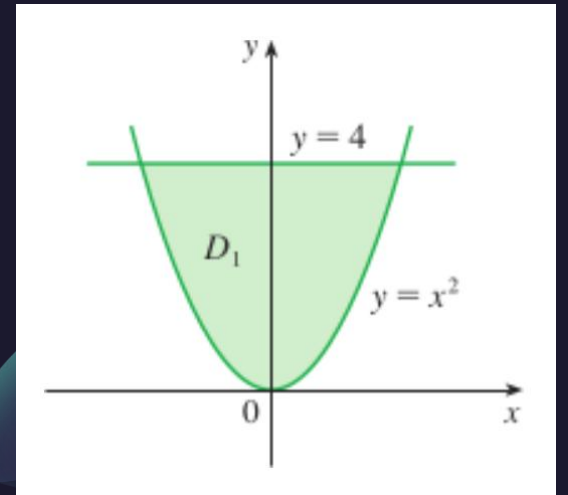
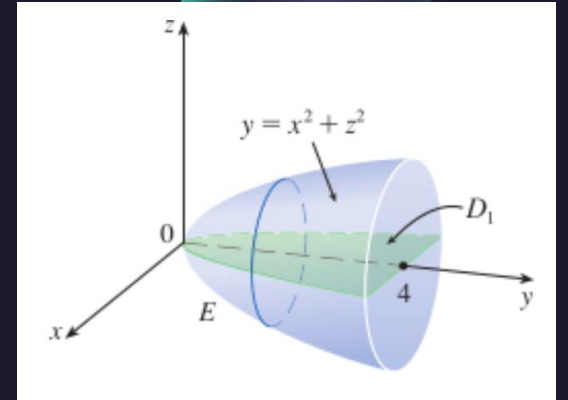
$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2+y^2) \, dz \, dy \, dx$$



Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2 + z^2} \, dV$$

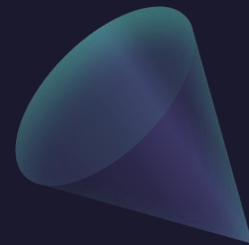
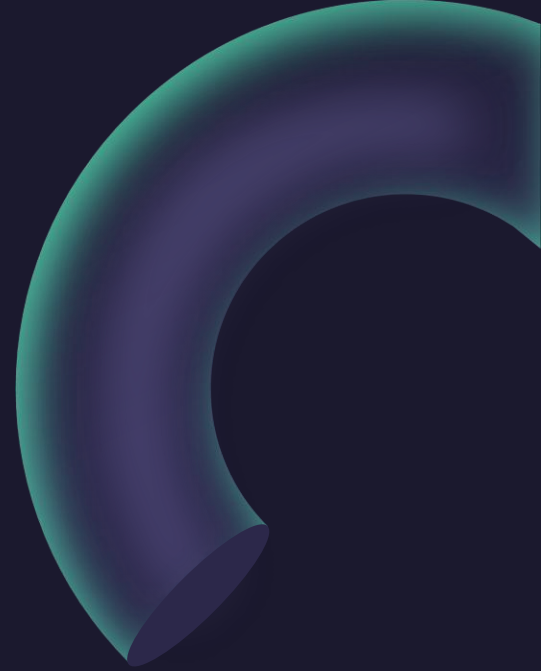
where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.



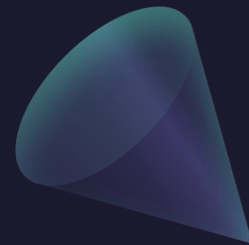
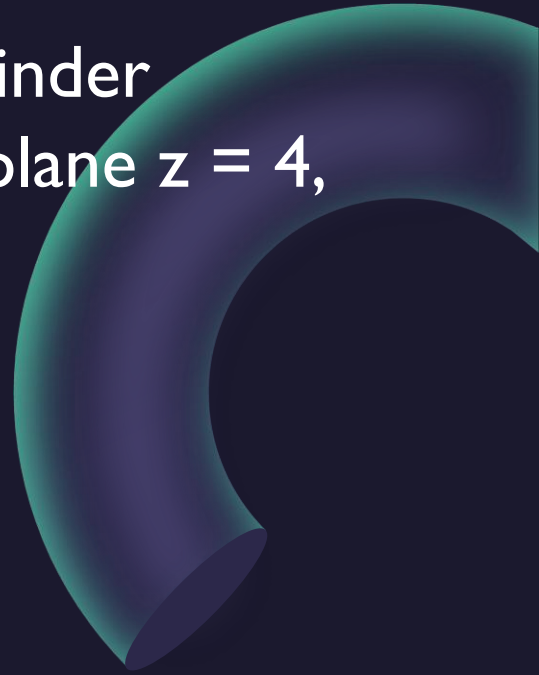
Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2 + z^2} \, dV$$

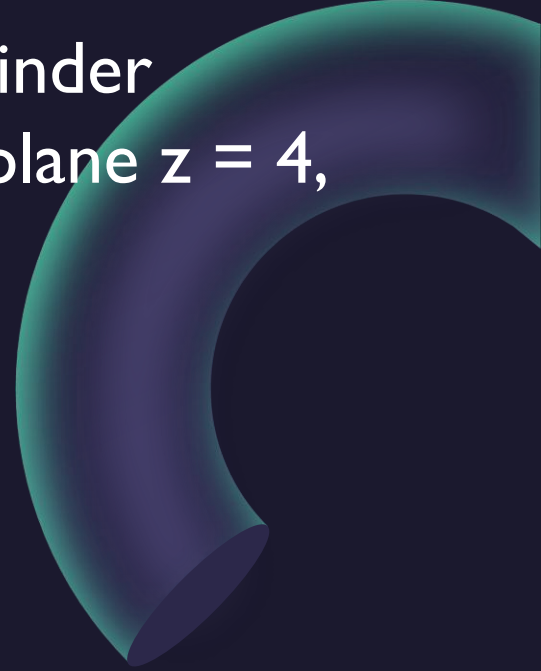
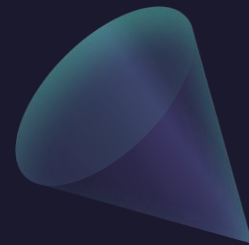
where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.



Example: Find the volume of the solid E bounded within the cylinder $x^2 + y^2 = 1$, to the right of the xz-plane, below the plane $z = 4$, and above $z = 1 - x^2 - y^2$.



Example: Find the volume of the solid E bounded within the cylinder $x^2 + y^2 = 1$, to the right of the xz-plane, below the plane $z = 4$, and above $z = 1 - x^2 - y^2$.



Thank you

Until next time.

